The Derivative from First Principles

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The definition of the derivative**

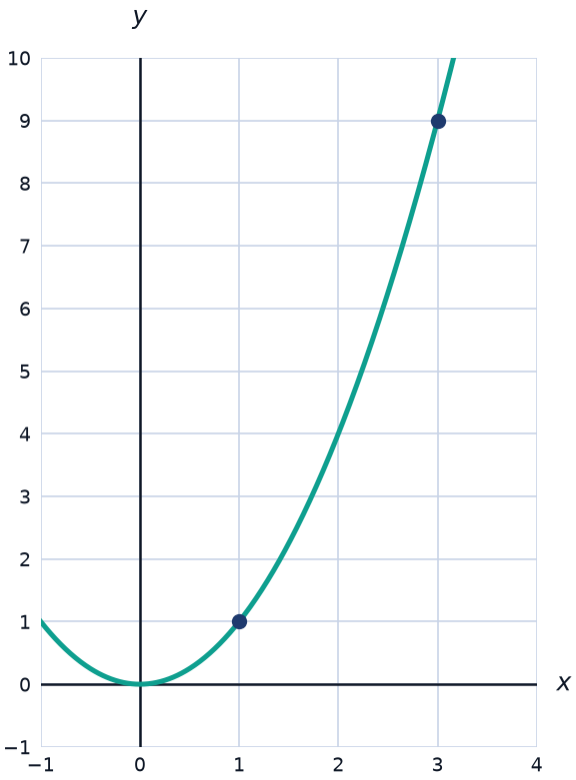
Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

2 **[Foundational]** • **Target: The definition of the derivative**

Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).

3 **[Foundational]** • **Target: The definition of the derivative**

The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.



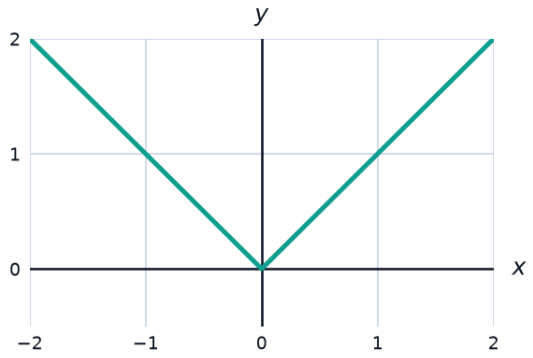
a Find the slope of the secant line.

b Find $f'(2)$ using the derivative, and compare it to the secant slope.

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4 [Foundational] • Target: Differentiability

Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.



5 [Foundational] • Target: Differentiability

State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

6 [Foundational] • Target: More derivatives from the definition

Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).

7 [Foundational] • Target: More derivatives from the definition

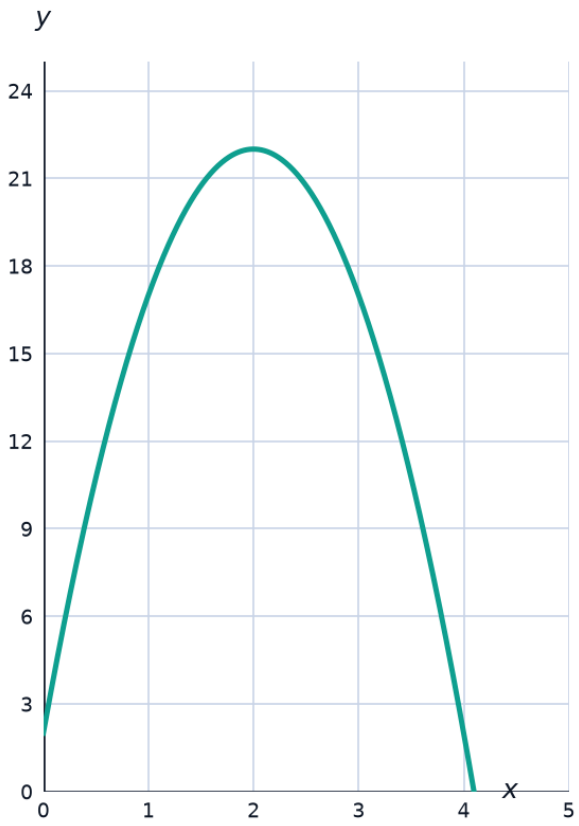
Use the limit definition to find $f'(x)$ for $f(x) = x^3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Interpreting the derivative**

A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.



9 **[Intermediate]** • **Target: Point-derivatives from the definition**

Use the limit definition $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(4)$ for $f(x) = x^2 - 3x$.

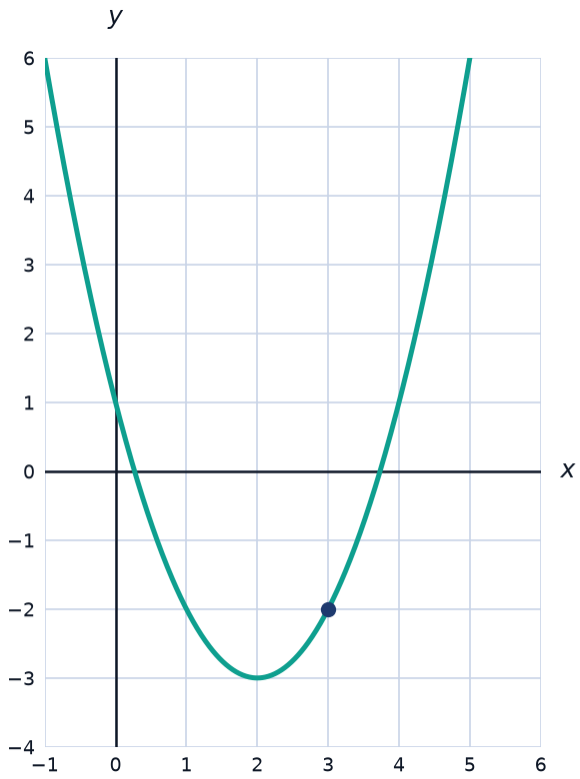
10 **[Intermediate]** • **Target: Point-derivatives from the definition**

Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

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11 **[Intermediate]** • **Target: The derivative as the slope of the tangent line**

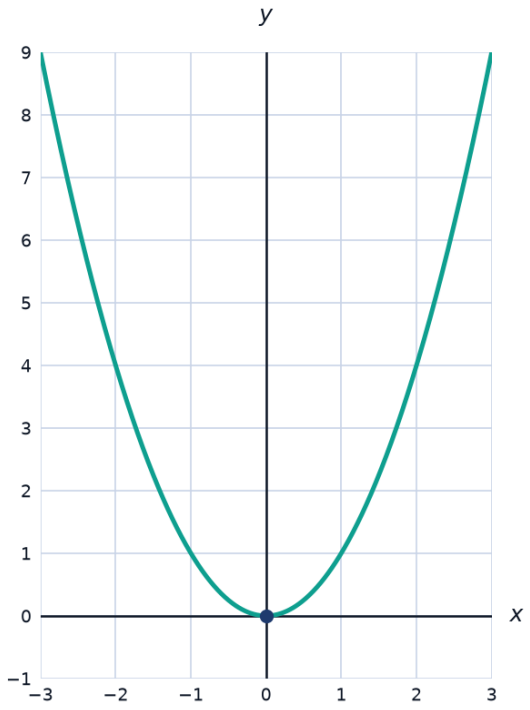
For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.



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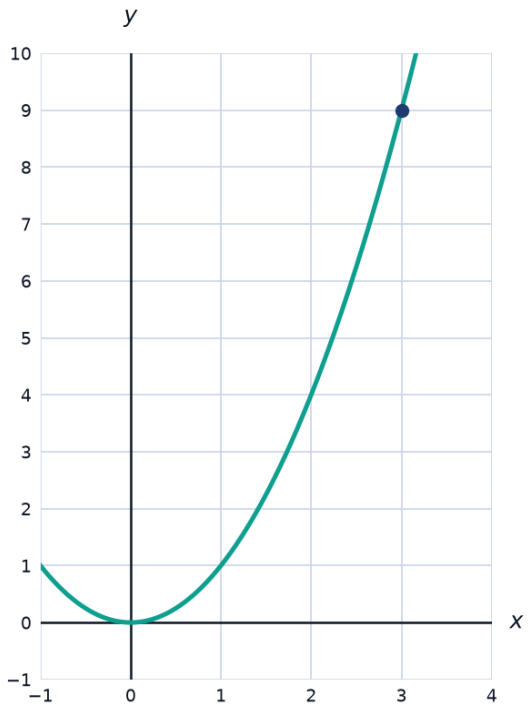
12 **[Intermediate]** • **Target: The derivative as the slope of the tangent line**

Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.



13 **[Intermediate]** • **Target: The derivative function versus the derivative at a point**

Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.



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14 [Intermediate] • Target: Estimating a derivative from a table

A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 11.01 Sequences and Series

Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

16 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 12.01 Convergence Tests

Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.

17 [Challenging] • Target: Interpreting the derivative • Cross-Lesson: 12.02 Power Series and Taylor Series

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$?

18 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Find the directional derivative of $f(x, y) = \sqrt{x^2 + y^2 + 4}$ at $(1,\ 3)$ in the direction of $\langle 4,\ -3 \rangle$.

19 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate $\int_0^{\pi} e^x \sin(2x)\, dx$.

20 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 15.01 Double Integrals

If $y = x^{\sin(x)}$, find $\frac{dy}{dx}$ at $x = 4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Limit Definition of the Derivative

State the standard limit definition of $f'(x)$.

a. $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

b. $f'(x) = \lim_{h \to 0} \frac{f(x) - f(h)}{h}$

c. $f'(x) = \frac{f(x+h) - f(x)}{h}$

d. $f'(x) = \lim_{x \to 0} \frac{f(x+h) - f(x)}{x}$

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22 [Foundational] • Target: The Limit Definition of the Derivative

Use first principles to find the derivative of $f(x) = 5x^2 - 3x$.

a. $10x - 3$

b. $10x$

c. $5x - 3$

d. $10x + 3$

23 [Foundational] • Target: The Limit Definition of the Derivative

Use the alternative limit definition $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

a. 12

b. 8

c. 6

d. 4

24 [Foundational] • Target: First Principles for Radicals and Reciprocals

Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$.

a. $-1/x^2$

b. $1/x^2$

c. $-1/x$

d. $-2/x^2$

25 [Foundational] • Target: First Principles for Radicals and Reciprocals

Find $f'(x)$ from first principles for $f(x) = \sqrt{x}$ $(x > 0)$.

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x}}$

c. $-\frac{1}{2\sqrt{x}}$

d. $\frac{2}{\sqrt{x}}$

26 [Foundational] • Target: First Principles for Radicals and Reciprocals

Evaluate $\lim_{h \to 0} \frac{(2+h)^5 - 32}{h}$ by recognizing it as a derivative.

a. 80

b. 40

c. 32

d. 16

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27 [Foundational] • Target: Differentiability and Continuity

Which statement correctly describes the relationship between differentiability and continuity?

a. Differentiability implies continuity, but continuity does not imply differentiability

b. Continuity implies differentiability

c. They are completely independent properties

d. A function is differentiable if and only if it is continuous

28 [Foundational] • Target: Differentiability and Continuity

Why does $f(x) = |x - 3|$ fail to be differentiable at $x = 3$?

a. The left-hand and right-hand derivative limits are unequal ($-1 \neq 1$)

b. $f(3)$ is undefined

c. f is discontinuous at $x = 3$

d. The tangent line is horizontal

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Differentiability and Continuity

At $x = 0$, the function $f(x) = x^{1/3}$ has which type of non-differentiability?

a. Vertical tangent ($f'(x) \to \infty$)

b. Corner / sharp turn

c. Cusp

d. Removable discontinuity

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30 [Intermediate] • Target: Geometric and Physical Interpretations

What geometric feature does the derivative $f'(a)$ represent?

a. The slope of the tangent line to $y = f(x)$ at $x = a$

b. The area under the curve from 0 to a

c. The secant line slope between 0 and a

d. The curvature at $(a, f(a))$

31 [Intermediate] • Target: Geometric and Physical Interpretations

A particle moves along a line with position $s(t) = 4t^2 - 3t + 1$. What is its instantaneous velocity at $t = 2$?

a. 13

b. 11

c. 16

d. 10

32 [Intermediate] • Target: Geometric and Physical Interpretations

Find the equation of the tangent line to $y = x^2 - 4x$ at the point where $x = 3$.

a. $y = 2x - 9$

b. $y = 2x - 3$

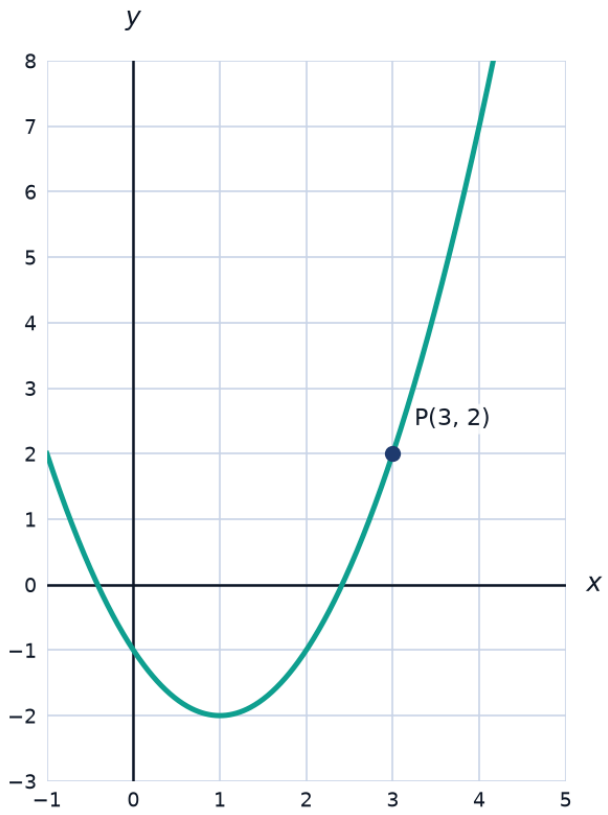
c. $y = -x$

d. $y = 4x - 15$

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33 [Intermediate] • Target: Geometric and Physical Interpretations

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



a. 4

b. 2

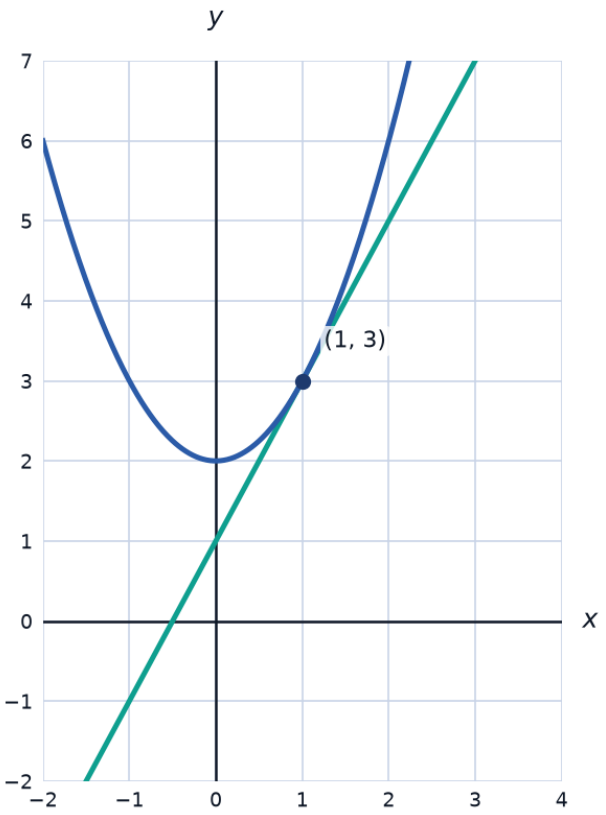
c. 3

d. 6

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34 **[Intermediate]** • **Target: The definition of the derivative**

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.



a. 1

b. 2

c. 3

d. 4

35 **[Intermediate]** • **Target: Differentiability**

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

a. 2

b. 4

c. 8

d. 10

36 **[Intermediate]** • **Target: More derivatives from the definition**

Tangent line slope to $y = \ln x$ at $x = e$:

a. 1

b. e

c. $1/e$

d. $\ln e$

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37 [Intermediate] • Target: Interpreting the derivative

The derivative of $f(x) = \frac{x^4 - 1}{x^2 + 1}$ at $x = 1$ using limit definition:

a. 1

b. 2

c. 3

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

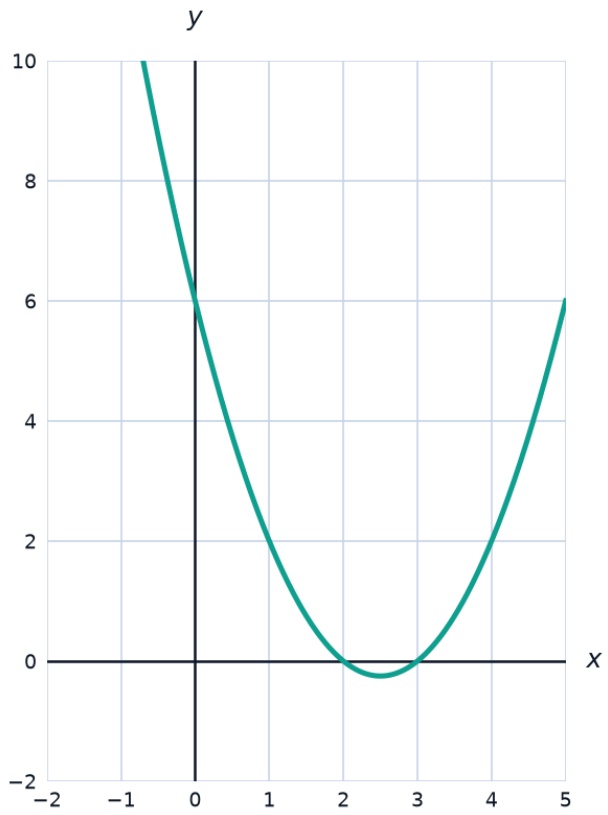
c. zero to zero

d. positive to negative

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39 **[Challenging]** • **Target: The derivative as the slope of the tangent line** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to x?



a. $2x - 5$

b. $x - 5$

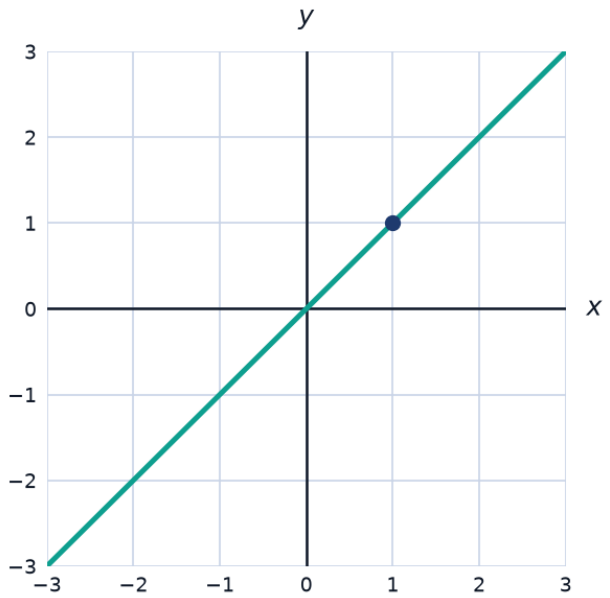
c. $x^2 - 5$

d. $2x$

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40 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 15.01 Double Integrals

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?



a. 1

b. 0

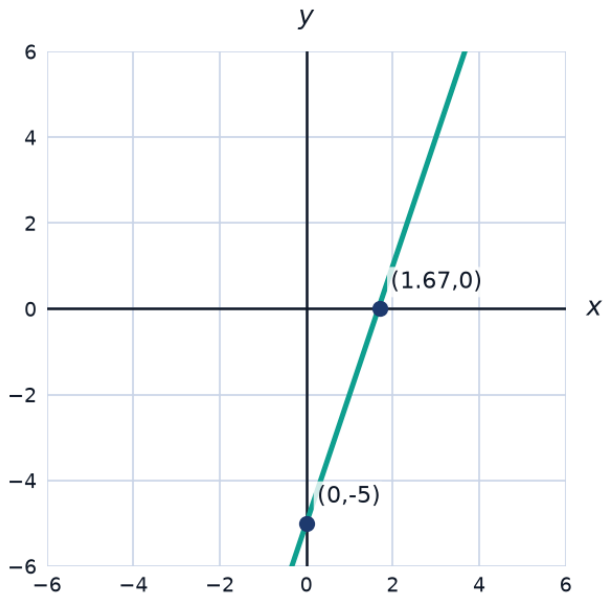
c. 2

d. Does not exist

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41 [Challenging] • Target: Estimating a derivative from a table • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The slope of $y = 3x - 5$ at any point is:



a. -5

b. 0

c. 3

d. undefined

42 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 2.01 Evaluating Limits

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

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43 [Challenging] • Target: The Limit Definition of the Derivative • Cross-Lesson: 2.02 Continuity

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

a. $2x - 3y$

b. $2x - 3xy + 4y$

c. $-3x + 4y$

d. $2x - 3y + 4y$

44 [Challenging] • Target: First Principles for Radicals and Reciprocals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Find the slope of the tangent line to $r = \sin(2\theta)$ at $\theta = \frac{\pi}{3}$.

a. -1

b. $-\frac{5\sqrt{3}}{3}$

c. $\frac{\sqrt{3}}{5}$

d. $\sqrt{3}$

45 [Challenging] • Target: Differentiability and Continuity • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Find the slope of the tangent line to $r = 2\sin(\theta) + 1$ at $\theta = \frac{2\pi}{3}$.

a. $-4 + 3\sqrt{3}$

b. -1

c. $-\frac{3\sqrt{3}}{11} - \frac{4}{11}$

d. $-\sqrt{3}$

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